Two-variable data: Models and scatterplots answer key

#	ID	Answer	Difficulty
1	363d5daa-100c-4c65-80bf-bd1ca694191d	B	Easy
2	41f938b0-0d18-48ac-b9ea-15cc947faaa2	C	Easy
3	01239-DC	C	Easy
4	0c969089-dec5-4180-9837-2658154630eb	B	Easy
5	77c322ee-efee-4f19-ad6f-03294924b08f	B	Easy
6	3a14beb2-aad4-426f-afd8-d1bd9dbaf599	C	Easy
7	9fe0c563-5c49-44d6-bd3d-3956ed2de2bd	C	Easy
8	07721-DC	B	Easy
9	76bc1af1-9e88-4760-961f-db2050ad9091	C	Easy
10	aee1cc54-9f2e-46de-94fa-2cccbe708d41	D	Easy
11	62f651c4-37e7-4fa9-8a3d-ff62ceb23066	C	Easy
12	026230-DC	A	Easy
13	027703-DC	B	Easy
14	3197f818-b46b-411b-ba07-d4d1c811fa29	B	Easy
15	070913-DC	B	Easy
16	070917-DC	?	Easy
17	69235d8f-a27b-49a1-a4e9-e905f8b73ef4	D	Easy
18	070915-DC	A	Easy
19	027702-DC	A	Easy
20	c15291f0-3442-4787-9e24-c2044f3e1f6f	B	Easy

21	4c22545c-b1dc-417e-a342-b8115c476ce0	B	Easy
22	07714-DC	D	Easy
23	070916-DC	B	Easy
24	bae655f1-291d-420a-881f-02c5d411251b	C	Easy
25	003efa6a-3fe0-4c5f-ac3d-7589e4ff53ee	40	Easy
26	be404513-86c1-473f-9573-d0ef7f6ac666	B	Easy
27	6401adca-8126-4887-bcb0-2e7076d6e999	A	Easy
28	031645-DC	C	Easy
29	f8342a9a-48d1-4fe3-865b-171f9d9f8f72	A	Easy
30	070914-DC	C	Easy
31	c00233af-cadc-465e-9aa2-54ba0b5e6923	C	Easy
32	234d1fe8-4e40-4728-ad3a-674eb0fb9188	A	Easy
33	031603-DC	B	Easy
34	dbe25324-030f-4284-a966-755078762e0c	B	Easy
35	027709-DC	A	Medium
36	57cffc23-e07d-4acc-9f42-b8eacfd50cd8	C	Medium
37	9b172c91-1885-4bc0-9faa-73425007e043	B	Medium
38	026150-DC	A	Medium
39	028872-DC	C	Medium
40	89491281-dcad-499e-8fe5-2563d680985f	C	Medium
41	11102-DC	?	Medium
42	08424-DC	B	Medium
43	031669-DC	C	Medium
44	71a3b42d-82b2-44f5-8a6d-50461b5cbd2f	0.5, 1/2	Medium
45	026242-DC	A	Medium
46	59d31da7-fdd1-4c5d-bf7e-1927db0fb130	B	Medium

47	f1d73552-b37d-4f00-870e-29249c5ffb4b	D	Medium
48	06aa14a7-9593-4ae2-9b7b-bf8575775587	A	Medium
49	a3419986-a630-4b5b-9c48-62a85f09b32e	D	Medium
50	741f638a-272e-4982-91a9-50a01668f587	C	Medium
51	76b9ab2a-a161-4577-8b1b-7e587c2a4543	A	Medium
52	028832-DC	D	Medium
53	44d16b54-4d25-47d8-9c06-f7e06ccbc36f	6	Medium
54	fdc411ff-1365-4fd5-a32d-1023475eb0e3	C	Medium
55	03f22abf-7da2-44d7-9e60-a131e0841a87	C	Medium
56	3804334d-fce7-4a1c-a320-674b9b7eab71	D	Medium
57	e798b8cb-3417-44f6-9cd9-077bb0117c35	C	Medium
58	86d0b8f2-f99e-4442-b537-5979e3167791	5	Medium
59	40d9d60a-1c6b-4ed3-82e1-5d956fe9153a	C	Medium
60	a83d8fcc-ddc5-4441-bd22-76b4ad2b5e1c	C	Medium
61	019531-DC	D	Hard
62	1862dad5-fe16-4d8c-8544-3cb91c1fd301	A	Hard
63	66189329-34b5-4940-a091-8d6219efb626	4.5, 9/2	Hard
64	87f878d4-305d-414f-8da8-f430f23c0bfd	D	Hard
65	08280-DC	C	Hard
66	14941-DC	B	Hard
67	11107-DC	A	Hard
68	020386-DC	A	Hard
69	070918-DC	?	Hard
70	10145-DC	D	Hard
71	3541703e-0b9d-494a-8a34-38ddbf560b9a	A	Hard

Rationales

▼ 1 — B

Choice B is correct. For the given line graph, the estimated number of chipmunks is represented on the vertical axis. The greatest estimated number of chipmunks in the state park is indicated by the greatest height in the line graph. This height is achieved when the year is 1994.

Choice A is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

▼ 2 — C

Choice C is correct. For the given line graph, the percent of cars for sale at a used car lot on a given day is represented on the vertical axis. The percent of cars for sale is the smallest when the height of the line graph is the lowest. The lowest height of the line graph occurs for cars with a model year of 2014.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

▼ 3 — C

Correct Answer Rationale
Choice C is correct. According to the model, the stopping distance, in feet, of a vehicle traveling 55 miles per hour is about 200 feet. Of the choices given, the best estimate of the stopping distance for a car traveling 55 miles per hour is 210 feet.

Incorrect Answer Rationale
Choices A, B, and D are incorrect and may be the result of incorrectly reading the given quadratic model. The corresponding x-values to the y-values of 25 and 30 are not part of the model. The corresponding x-value to a y-value of 250 is approximately 60 mph, not 55 mph.

▼ 4 — B

Choice B is correct. The equation representing a linear model can be written in the form $y = a + bx$, or $y = bx + a$, where b is the slope of the graph of the model and $0, a$ is the y-intercept of the graph of the model. The scatterplot shows that as the x-values of the data points increase, the y-values of the data points decrease, which means the graph of an appropriate linear model has a negative slope. Therefore, $b < 0$. The scatterplot also shows that the data points are close to the y-axis at a positive value of y. Therefore, the y-intercept of the graph of an appropriate linear model has a positive y-coordinate, which means $a > 0$. Of the given choices, only choice B, $y = -1.9x + 10.1$, has a negative value for b, the slope, and a positive value for a, the y-coordinate of the y-intercept.

Choice A is incorrect. The graph of this model has a y-intercept with a negative y-coordinate, not a positive y-coordinate.

Choice C is incorrect. The graph of this model has a positive slope, not a negative slope, and a y-intercept with a negative y-coordinate, not a positive y-coordinate.

Choice D is incorrect. The graph of this model has a positive slope, not a negative slope.

▼ 5 — B

Choice B is correct. The line of best fit shown intersects the y-axis at a positive y-value and has a negative slope. The graph of an equation of the form $y = a + bx$, where a and b are constants, intersects the y-axis at a y-value of a and has a slope of b. Of the given choices, only choice B represents a line that intersects the y-axis at a positive y-value, 13.5, and has a negative slope, -0.8.

Choice A is incorrect. This equation represents a line that has a positive slope, not a negative slope.

Choice C is incorrect. This equation represents a line that intersects the y-axis at a negative y-value, not a positive y-value, and has a positive slope, not a negative slope.

Choice D is incorrect. This equation represents a line that intersects the y-axis at a negative y-value, not a positive y-value.

▼ 6 — C

Choice C is correct. For each point on the scatterplot shown, the x-value represents the weight, in pounds, of a female gray wolf and the y-value represents the number of offspring

that wolf produced. The point on the graph with an x-value of 50 has a y-value of 6. Therefore, the 50-pound gray wolf produced 6 offspring.

Choice A is incorrect. One of the wolves produced 8 offspring, but its weight was greater than 50 pounds.

Choice B is incorrect. Three of the wolves produced 7 offspring each, but their weights were each greater than 50 pounds.

Choice D is incorrect. Two of the wolves produced 5 offspring each, but their weights were each less than 50 pounds.

▼ 7 — C

Choice C is correct. A linear model can be written in the form $y = mx + b$, where m is the slope of the graph of the model in the xy-plane and $(0, b)$ is the y-intercept. The graph of an appropriate linear model for the data shown in the scatterplot passes near the points $(0, 0)$ and $(4, 2.4)$ in the xy-plane. Two points on a line, x_1, y_1 and x_2, y_2, can be used to find the slope of the line using the slope formula, $m = \frac{y_2 - y_1}{x_2 - x_1}$. Substituting the points $(0, 0)$ and $(4, 2.4)$ for x_1, y_1 and x_2, y_2, respectively, in the slope formula yields $m = \frac{2.4 - 0}{4 - 0}$, or $m = 0.60$. Therefore, the value of m for an appropriate linear model is approximately 0.60. Substituting 0.60 for m in $y = mx + b$ yields $y = 0.60x + b$. Since an appropriate linear model passes near the point $(0, 0)$, it follows that b is approximately 0 and an appropriate linear model for the data is $y = 0.60x + 0$. Thus, of the given choices, $y = 0.606x$ is the most appropriate linear model for these data.

Alternate approach: A linear model can be written in the form $y = mx + b$, where m is the slope of the graph of the model in the xy-plane and $(0, b)$ is the y-intercept. The scatterplot shows that as the x-values of the data points increase, the y-values of the data points increase, which means the graph of an appropriate linear model has a positive slope. The scatterplot also shows that an appropriate linear model would pass close to the point $(0, 0)$. Of the given choices, $y = 0.606x$ is the only linear model whose graph has a positive slope and a y-intercept of $(0, 0)$.

Choice A is incorrect. The graph of this model has a negative slope, not a positive slope.

Choice B is incorrect. The graph of this model has a negative slope, not a positive slope, and a y-intercept that is not $(0, 0)$.

Choice D is incorrect. The graph of this model has a slope that is too steep to fit the data, and a y-intercept that is not $(0, 0)$.

▼ 8 — B

Choice B is correct. Theresa's speed was increasing from 0 to 5 minutes and from 20 to 25 minutes, which is a total of 10 minutes. Theresa's speed was decreasing from 10 minutes to 20 minutes and from 25 to 30 minutes, which is a total of 15 minutes. Therefore, Theresa's speed was NOT increasing for a longer period of time than it was decreasing.

Choice A is incorrect. Theresa ran at a constant speed for the 5-minute period from 5 to 10 minutes. Choice C is incorrect. Theresa's speed decreased at a constant rate during the last 5 minutes, which can be seen since the graph is linear during that time. Choice D is incorrect. Theresa's speed reached its maximum at 25 minutes, which is within the last 10 minutes.

▼ 9 — C

Choice C is correct. For the line graph shown, the probability of snow, as a percent, is represented on the vertical axis. According to the line graph, during this four-day period, the probability of snow is 30% for Thursday.

Choice A is incorrect. The probability of snow on Tuesday is 60%.

Choice B is incorrect. The probability of snow on Wednesday is 90%.

Choice D is incorrect. The probability of snow on Friday is 70%.

▼ 10 — D

Choice D is correct. An appropriate model should follow the trend of the data points and should have data points both above and below the model. The scatterplot shows that the data points have an increasing trend that is curved. Therefore, an appropriate model should be an increasing curve with data points both above and below the model. Of the given choices, only the model in choice D is an increasing curve with data points both above and below the model.

Choice A is incorrect. Since the trend of the data points isn't linear, a line isn't the most appropriate model for the data.

Choice B is incorrect. Since the trend of the data points is increasing and isn't linear, a decreasing line isn't the most appropriate model for the data.

Choice C is incorrect. All the data points are below the model shown in this graph.

▼ 11 — C

Choice C is correct. It's given that the scatterplot shows the relationship between two variables, *x* and *y*, and a line of best fit is shown. For the line of best fit shown, for each increase in the value of *x* by 1, the corresponding value of *y* increases by a constant rate. It follows that the relationship between the variables *x* and *y* has a positive linear trend. A line in the *xy*-plane that passes through the points a, b and c, d has a slope of $\frac{d-b}{c-a}$. The line of best fit shown passes approximately through the points $0, 0.25$ and $4, 2$. It follows that the slope of this line is approximately $\frac{2-0.25}{4-0}$, which is equivalent to 0.4375. Therefore, of the given choices, 0.44 is closest to the slope of the line of best fit shown.

Choice A is incorrect. This is the slope of a line of best fit for a relationship between *x* and *y* that has a negative, rather than a positive, linear trend.

Choice B is incorrect. This is the slope of a line of best fit for a relationship between *x* and *y* that has a negative, rather than a positive, linear trend.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 12 — A

Choice A is correct. The year with the fewest graduates who enrolled in college within 24 months of graduation is the point with the lowest value on the vertical axis. This occurs at 2002.

Choice B, C, and D are incorrect. The years 2004, 2005, and 2007 each had a greater number of graduates who enrolled in college within 24 months of graduation than did the year 2002.

▼ 13 — B

Choice B is correct. The graphical model that most closely fits the data in the scatterplot is a model in which the number of data points above and below the model are approximately balanced. Fitting a graphical model to the data shown results in an upward-facing parabola with a y-intercept near $(0,20)$ and a vertex with an approximate x-value of 2.5. Of the given

choices, only choice B gives an equation of an upward-facing parabola with a y-intercept at $(0,20)$. Furthermore, substituting 2.5 for x into the equation in choice B yields $y=5$. This is approximately the y-value of the vertex of the model.

Choices A, C, and D are incorrect. These equations don't give a graphical model that best fits the data. At $x=0$, they have y-values of -20, -3, and 3, respectively. At $x=2.5$, they have y-values of -35, -3, and 3, respectively.

▼ 14 — B

Choice B is correct. The graph of function *f* shows that as *x* increases, *f*(*x*) also increases, which means that *f* is an increasing function. The graph of function *f* is curved upward and increases at an increasing rate, not a constant rate. This behavior is characteristic of exponential functions, not linear functions. Therefore, of the given choices, function *f* is best described as increasing exponential.

Choice A is incorrect. For a decreasing function, as *x* increases, *f*(*x*) decreases rather than increases.

Choice C is incorrect. For a decreasing function, as *x* increases, *f*(*x*) decreases rather than increases, and the graph of a linear function isn't curved.

Choice D is incorrect. The graph of a linear function isn't curved.

▼ 15 — B

Choice B is correct. The association between the elevation and the temperature of locations in the Lake Tahoe Basin can be described by looking at the direction of the line of best fit. The line of best fit slopes downward, which corresponds to the temperature decreasing as the elevation increases.

Choices A and C are incorrect. Both of these choices would be represented by a line of best fit that slopes from the lower left to the upper right of the graph, which isn't what's shown on the graph. Choice D is incorrect. This choice would be represented by a line of best fit that is horizontal or has a slope very close to 0. This is not what's shown on the graph.

▼ 16 — ?

The correct answer is 480. An average temperature of $32°C$ corresponds to the value 32 on the x-axis. On the line of best fit, an x-value of 32 corresponds to a y-value of 480. The values on the y-axis correspond to the number of people predicted to visit this beach. Therefore, 480 people are predicted to visit this beach on a day with an average temperature of $32°C$.

▼ 17 — D

Choice D is correct. A linear model can be written in the form $y = mx + b$, where m is the slope of the graph of the model in the xy-plane and $(0, b)$ is the y-intercept. The graph of an appropriate linear model for this relationship passes near the points $(1, 3)$ and $(9, 10)$ in the xy-plane. Two points on a line, x_1, y_1 and x_2, y_2, can be used to find the slope of the line using the slope formula, $m = \frac{y_2 - y_1}{x_2 - x_1}$. Substituting the points $(1, 3)$ and $(9, 10)$ for x_1, y_1 and x_2, y_2, respectively, in the slope formula yields $m = \frac{10 - 3}{9 - 1}$, or $m = 0.875$. Therefore, the value of m for an appropriate linear model is approximately 0.875. Substituting 0.875 for m in $y = mx + b$ yields $y = 0.875x + b$. Since an appropriate linear model passes near the point $(1, 3)$, the approximate value of b can be found by substituting 1 for x and 3 for y in the equation $y = 0.875x + b$, which yields $3 = (0.875)(1) + b$, or $3 = 0.875 + b$. Subtracting 0.875 from both sides of this equation yields $2.125 = b$. Therefore, the value of b for an appropriate linear model is approximately 2.125. Thus, of the given choices, $y = 0.9x + 2.2$ is the most appropriate linear model for this relationship.

Alternate approach: A linear model can be written in the form $y = mx + b$, where m is the slope of the graph of the model in the xy-plane and $(0, b)$ is the y-intercept. The scatterplot shows that as the x-values of the data points increase, the y-values of the data points increase, which means the graph of an appropriate linear model has a positive slope. Of the given choices, $y = 0.9x + 2.2$ is the only linear model whose graph has a positive slope.

Choice A is incorrect. The graph of this model has a negative slope, not a positive slope.

Choice B is incorrect. The graph of this model has a negative slope, not a positive slope.

Choice C is incorrect. The graph of this model has a negative slope, not a positive slope.

▼ 18 — A

Choice A is correct. The average rate of change in temperature of the coffee in degrees Fahrenheit per minute is calculated by dividing the difference between two recorded temperatures by the number of minutes in the corresponding interval of time. Since the time

intervals given are all 10 minutes, the average rate of change is greatest for the points with the greatest difference in temperature. Of the choices, the greatest difference in temperature occurs between 0 and 10 minutes.

Choices B, C, and D are incorrect and may result from misinterpreting the average rate of change from the graph.

▼ 19 — A

Choice A is correct. A line of best fit for the data in a scatterplot is a line that follows the trend of the data with approximately half the data points above and half the data points below the line. Based on the given data, a line of best fit will have a positive y-intercept on or near the point $(0,6)$ and a negative slope. All of the choices are in slope-intercept form $y = mx + b$, where m is the slope and b is the y-coordinate of the y-intercept. Only choice A is an equation of a line with a positive y-intercept at $(0,6)$ and a negative slope, -1.

Choice B is incorrect. This equation is for a line that has a negative y-intercept, not a positive y-intercept. Choices C and D are incorrect and may result from one or more sign errors and from switching the values of the y-intercept and the slope in the equation.

▼ 20 — B

Choice B is correct. It's given that the airplane descends at a constant rate of $400 \text{ feet per minute}$. Since the altitude decreases by a constant amount during each fixed time period, the relationship between the airplane's altitude and time is linear. Since the airplane descends from an altitude of $9{,}500 \text{ feet}$ to $5{,}000 \text{ feet}$, the airplane's altitude is decreasing with time. Thus, the relationship is best modeled by a decreasing linear function.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 21 — B

Choice B is correct. The given values show that as x increases, fx also increases, which means that f is an increasing function. Furthermore, fx increases at a constant rate of 1 for

each increase of x by 1. A function with a constant rate of change is linear. Thus, the function f can be described as an increasing linear function.

Choice A is incorrect. For a decreasing linear function, as x increases, fx decreases rather than increases.

Choice C is incorrect. For a decreasing exponential function, for each increase of x by 1, fx decreases by a fixed percentage rather than increases at a constant rate.

Choice D is incorrect. For an increasing exponential function, for each increase of x by 1, fx increases by a fixed percentage rather than at a constant rate.

▼ 22 — D

Choice D is correct. The change in the number of 3-D movies released between any two consecutive years can be found by first estimating the number of 3-D movies released for each of the two years and then finding the positive difference between these two estimates. Between 2003 and 2004, this change is approximately $2-2=0$ movies; between 2008 and 2009, this change is approximately $20-8=12$ movies; between 2009 and 2010, this change is approximately $26-20=6$ movies; and between 2010 and 2011, this change is approximately $46-26=20$ movies. Therefore, of the pairs of consecutive years in the choices, the greatest increase in the number of 3-D movies released occurred during the time period between 2010 and 2011.

Choices A, B, and C are incorrect. Between 2010 and 2011, approximately 20 more 3-D movies were released. The change in the number of 3-D movies released between any of the other pairs of consecutive years is significantly smaller than 20.

▼ 23 — B

Choice B is correct. The line of best fit passes through the point $(8{,}500,\ 39)$. Therefore, the line of best fit predicts a temperature of 39°F for a location in Lake Tahoe Basin with an elevation of 8,500 feet.

Choice A is incorrect. This is the lowest temperature listed on the scatterplot, and the line of best fit never crosses this value for any of the elevations shown. Choice C is incorrect. According to the line of best fit, the temperature of 41°F is predicted for an elevation of slightly greater than 7,500 feet, not an elevation of 8,500 feet. Choice D is incorrect.

According to the line of best fit, the temperature of 43°F is predicted for an elevation of roughly 6,700 feet, not an elevation of 8,500 feet.

▼ 24 — C

Choice C is correct. On the line of best fit, an x-value of $1{,}200$ corresponds to a y-value between 10 and 12. Therefore, of the given choices, 11 is closest to the y-value predicted by the line of best fit at $x = 1{,}200$.

Choice A is incorrect. This is the integer value closest to the y-value predicted by the line of best fit at $x = 1{,}800$.

Choice B is incorrect. This is the integer value closest to the y-value predicted by the line of best fit at $x = 1{,}500$.

Choice D is incorrect. This is the integer value closest to the y-value predicted by the line of best fit at $x = 600$.

▼ 25 — 40

The correct answer is 40. For each data point on the scatterplot, the x-value represents the growing season after transplantation and the y-value represents the weight, in pounds, of all nectarines that grew on the tree during the season. The scatterplot shows a data point at $4, 40$. It follows that during the 4th growing season after the tree's transplantation, 40 pounds of nectarines grew on the tree.

▼ 26 — B

Choice B is correct. In the given scatterplot, the x-values represent the distance above sea level, in feet, and the y-values represent the temperature, in $°\text{F}$. The point on the line of best fit with an x-value of $4{,}000$ has a corresponding y-value of 35. Therefore, at a distance of $4{,}000$ feet above sea level, the temperature predicted by the line of best fit is $35°\text{F}$.

Choice A is incorrect. This is the temperature, in $°\text{F}$, predicted by the line of best fit at a distance of 0 feet above sea level.

Choice C is incorrect. This is the measured temperature, in $°\text{F}$, at a distance of $6{,}000$ feet above sea level.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 27 — A

Choice A is correct. The line of best fit shown has a positive slope and intersects the y-axis at a positive y-value. The graph of an equation of the form $y = mx + b$, where m and b are constants, has a slope of m and intersects the y-axis at a y-value of b. Of the given choices, only $y = x + 3.4$ represents a line that has a positive slope, 1, and intersects the y-axis at a positive y-value, 3.4.

Choice B is incorrect. This equation represents a line that intersects the y-axis at a negative y-value, not a positive y-value.

Choice C is incorrect. This equation represents a line that has a negative slope, not a positive slope.

Choice D is incorrect. This equation represents a line that has a negative slope, not a positive slope, and intersects the y-axis at a negative y-value, not a positive y-value.

▼ 28 — C

Choice C is correct. The data in the scatterplot show an increasing linear trend. The density when the juice concentration is 60% will be between the densities shown at about 53% and 67% concentration, or between about 1,255 and 1,340 kg/m^3. Of the choices given, only 1,300 falls within this range.

Choices A, B, and D are incorrect. These are the approximate densities of grape juice with a concentration of 45%, 55%, and 70%, respectively.

▼ 29 — A

Choice A is correct. The graph of function f shows that as x increases, fx also increases, which means fx is an increasing function. The graph of f is a line, which indicates a constant rate of change. A function that has a constant rate of change is a linear function. Therefore, function f can be described as increasing linear.

Choice B is incorrect. For a decreasing function, as x increases, fx decreases, rather than increases.

Choice C is incorrect. The graph of an exponential function isn't a line.

Choice D is incorrect. For a decreasing function, as x increases, $f(x)$ decreases, rather than increases, and the graph of an exponential function isn't a line.

▼ 30 — C

Choice C is correct. According to the line of best fit, a planetoid with a distance from the Sun of 1.2 AU has a predicted density between $4.5\ \text{g/cm}^3$ and $4.75\ \text{g/cm}^3$. The only choice in this range is 4.6.

Choices A, B, and D are incorrect and may result from misreading the information in the scatterplot.

▼ 31 — C

Choice C is correct. The scatterplot shows that there was an increase in recorded temperature from $x = 2$ to $x = 3$ and from $x = 6$ to $x = 7$. When $x = 2$, the recorded temperature was approximately $60°\text{F}$ and when $x = 3$, the recorded temperature was greater than $70°\text{F}$. This means that the increase in recorded temperature from $x = 2$ to $x = 3$ was greater than $(70 - 60)°\text{F}$, or $10°\text{F}$. When $x = 6$, the recorded temperature was greater than $60°\text{F}$ and when $x = 7$, the recorded temperature was less than $70°\text{F}$. This means that the increase in recorded temperature from $x = 6$ to $x = 7$ was less than $(70 - 60)°\text{F}$, or $10°\text{F}$. It follows that the greatest increase in recorded temperature took place from $x = 2$ to $x = 3$.

Choice A is incorrect. The increase in recorded temperature from $x = 6$ to $x = 7$ was less than the increase in recorded temperature from $x = 2$ to $x = 3$.

Choice B is incorrect. From $x = 5$ to $x = 6$, a decrease, not an increase, in recorded temperature took place.

Choice D is incorrect. From $x = 1$ to $x = 2$, a decrease, not an increase, in recorded temperature took place.

▼ 32 — A

Choice A is correct. The line of best fit shown intersects the y-axis at a positive y-value and has a positive slope. The graph of an equation of the form $y = a + bx$, where a and b are constants, intersects the y-axis at a y-value of a and has a slope of b. Of the given choices, only choice A represents a line that intersects the y-axis at a positive y-value, 2.8, and has a positive slope, 1.7.

Choice B is incorrect. This equation represents a line that has a negative slope, not a positive slope.

Choice C is incorrect. This equation represents a line that intersects the y-axis at a negative y-value, not a positive y-value.

Choice D is incorrect. This equation represents a line that intersects the y-axis at a negative y-value, not a positive y-value, and has a negative slope, not a positive slope.

▼ 33 — B

Choice B is correct. According to the graph, when the car is traveling at 50 miles per hour, the braking distance is approximately 225 feet, and when the car is traveling at 30 miles per hour, the braking distance is approximately 100 feet. The difference between these braking distances is $225 - 100$, or 125 feet.

Choice A is incorrect and may result from finding the braking distance for 20 miles per hour, the difference between the given speeds. Choice C is incorrect and may result from subtracting the speed from the braking distance at 50 miles per hour. Choice D is incorrect and may result from finding the difference in the braking distances at 60 and 20 miles per hour.

▼ 34 — B

Choice B is correct. It's given that the scatterplot shows the relationship between the number of swans, y, and the number of years since 1970, x, where $0 \leq x \leq 50$. According to the scatterplot, the number of swans increases as the number of years since 1970 increases, and the data shows an exponential growth pattern rather than a linear pattern. When $a > 0$ and $b > 1$, an exponential model of the form $y = a(b)^x$ represents exponential growth and a model of the form $y = ab^{-x}$ represents exponential decay. The models in all four choices show $a > 0$ and $b > 1$. Choices B and D show models of the form $y = a(b)^x$ and thus represent exponential growth. The initial value a should approximate the y-value when $x = 0$, which is a little more than 40, so $a = 43$ is reasonable and consistent with the graph. Therefore, the equation $y = 43(1.04)^x$ is the most appropriate exponential model for the data.

Choice A is incorrect. The model $y = 43(1.04)^{-x}$ represents exponential decay, rather than exponential growth, which doesn't match the pattern in the scatterplot.

Choice C is incorrect. The model $y = 288(1.04)^{-x}$ represents exponential decay rather than exponential growth, which doesn't match the pattern in the scatterplot.

Choice D is incorrect. The model $y = 288(1.04)^x$ has an initial value of 288 when $x = 0$, so the initial value doesn't match the scatterplot.

▼ 35 — A

Choice A is correct. The data point with $x = 1$ has a y-coordinate of 12. The y-value predicted by the line of best fit at $x = 1$ is approximately 11. The difference between the y-coordinate of the data point and the y-value predicted by the line of best fit at $x = 1$ is $12 - 11$, or 1.

Choices B and C are incorrect and may result from incorrectly reading the scatterplot. Choice D is incorrect. This is the y-coordinate of the data point at $x = 1$.

▼ 36 — C

Choice C is correct. A line of best fit is shown in the scatterplot such that as the value of x increases, the value of y decreases. It follows that the slope of the line of best fit shown is negative. The slope of a line in the xy-plane that passes through the points x_1, y_1 and x_2, y_2 can be calculated as $\frac{y_2 - y_1}{x_2 - x_1}$. The line of best fit shown passes approximately through the points $0, 8$ and $10, 1$. Substituting $0, 8$ for x_1, y_1 and $10, 1$ for x_2, y_2 in $\frac{y_2 - y_1}{x_2 - x_1}$ yields the slope of the line being approximately $\frac{1 - 8}{10 - 0}$, which is equivalent to $\frac{-7}{10}$, or -0.7. Therefore, of the given choices, -0.7 is the closest to the slope of this line of best fit.

Choice A is incorrect. The line of best fit shown has a negative slope, not a positive slope.

Choice B is incorrect. The line of best fit shown has a negative slope, not a positive slope.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 37 — B

Choice B is correct. A line of best fit is shown in the scatterplot such that as the value of x increases, the value of y decreases. Thus, the slope of the line of best fit shown is negative. The slope of a line passing through two points, x_1, y_1 and x_2, y_2, can be calculated as $\frac{y_2 - y_1}{x_2 - x_1}$. The line of best fit shown passes approximately through the points $1, 12$ and $11, 4$. Substituting $1, 12$ and $11, 4$ for x_1, y_1 and x_2, y_2, respectively, in $\frac{y_2 - y_1}{x_2 - x_1}$ gives $\frac{4 - 12}{11 - 1}$, which is equivalent to $-\frac{8}{10}$, or -0.8. Therefore, of the given choices, -0.8 is closest to the slope of the line of best fit shown.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. The line of best fit shown has a negative slope, not a positive slope.

Choice D is incorrect. The line of best fit shown has a negative slope, not a positive slope.

▼ 38 — A

Choice A is correct. An equation of a line of best fit can be written in the form $y = a + bx$, where a is the y-intercept of the line and b is the slope. In the scatterplot shown, the line of best fit intersects the y-axis just over halfway between 10,000 and 20,000, or approximately 16,000. The line of best fit also intersects the graph at $(5,40,000)$. Using the slope formula $b = \frac{y_2 - y_1}{x_2 - x_1}$ and two points that lie on the graph such as $(5,40,000)$ and $(0,16,000)$, the slope can be approximated as $\frac{40,000 - 16,000}{5 - 0}$, or 4,800. Only choice A has a y-intercept near the estimate of 16,000 and a slope near the estimate of 4,800. Therefore, an equation of the line of best fit could be $n = 16,090 + 4,680t$.

Choice B is incorrect because the values for the slope and the y-coordinate of the y-intercept are switched. Choice C is incorrect because the value for the slope is approximately double the actual slope. Choice D is incorrect because the values for the slope and the y-intercept are switched and because the slope is approximately double the actual slope.

▼ 39 — C

Choice C is correct. A line that most closely fits the data is a line with an approximately balanced number of data points above and below the line. Fitting a line to the data shown results in a line with an approximate slope of 3 and a y-intercept near the point $(0,1)$. An equation for the line can be written in slope-intercept form, $y = mx + b$, where m is the slope and b is the y-coordinate of the y-intercept. The equation $y = 3.30x + 0.82$ in choice C fits the data most closely.

Choices A and B are incorrect because the slope of the lines of these equations is 0.82, which is a value that is too small to be the slope of the line that fits the data shown. Choice D is incorrect. The graph of this equation has a y-intercept at $(0,-3.30)$, not $(0,0.82)$. This line would lie below all of the data points, and therefore would not closely fit the data.

▼ 40 — C

Choice C is correct. A line in the *xy*-plane that passes through points x_1, y_1 and x_2, y_2 has a slope of $\frac{y_2 - y_1}{x_2 - x_1}$. The line of best fit shown passes approximately through the points $0, 0.2$ and $5, 9.3$. It follows that the slope of this line is approximately $\frac{9.3 - 0.2}{5 - 0}$, which is equivalent to $\frac{9.1}{5}$, or 1.82. Therefore, of the given choices, 1.8 is closest to the slope of the line of best fit shown.

Choice A is incorrect. This value is closest to the *y*-coordinate of the *y*-intercept of the line of best fit shown.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 41 — ?

Choice B is correct. Since the work is done at a constant rate, a linear model best models the situation. The number of shirts remaining is dependent on the length of time the inspector has worked; therefore, if the relationship were graphed, time would be the variable of the horizontal axis and the number of remaining shirts would be the variable of the vertical axis. Since the number of shirts decreases as the time worked increases, it follows that the slope of this graph is negative.

Choice A is incorrect and may result from incorrectly reasoning about the slope. Choices C and D are incorrect and may result from not identifying the constant rate of work as a characteristic of a linear model.

▼ 42 — B

Choice B is correct. The charity with the greatest percent of total expenses spent on programs is represented by the highest point on the scatterplot; this is the point that has a vertical coordinate slightly less than halfway between 90 and 95 and a horizontal coordinate slightly less than halfway between 3,000 and 4,000. Thus, the charity represented by this point has a total income of about $3,400 million and spends about 92% of its total expenses on programs. The percent predicted by the line of best fit is the vertical coordinate of the point on the line of best fit with horizontal coordinate $3,400 million; this vertical coordinate is very slightly more than 85. Thus, the line of best fit predicts that the charity with the greatest percent of total expenses spent on programs will spend slightly more than 85% on programs. Therefore, the difference between the actual percent (92%) and the prediction (slightly more than 85%) is slightly less than 7%.

Choice A is incorrect. There is no charity represented in the scatterplot for which the difference between the actual percent of total expenses spent on programs and the percent predicted by the line of best fit is as much as 10%. Choices C and D are incorrect. These choices may result from misidentifying in the scatterplot the point that represents the charity with the greatest percent of total expenses spent on programs.

▼ 43 — C

Choice C is correct. At $x = 0$, the value of 2^x is less than the value of $2x + 2$: $2^0 < 2(0) + 2$, which is equivalent to $1 < 2$. As the value of x increases, the value of 2^x remains less than the value of $2x + 2$ until $x = 3$, which is when the two values are equal: $2^3 = 2(3) + 2$, which is equivalent to $8 = 8$. Then, for $x > 3$, the value of 2^x is greater than the value of $2x + 2$. So there is a constant, 3, such that when $0 < x < 3$, then $2^x < 2x + 2$, but when $x > 3$, then $2^x > 2x + 2$.

Choice A is incorrect because $2^x > 2x + 2$ when $x > 3$. Choice B is incorrect because $2^x < 2x + 2$ when $0 < x < 3$. Choice D is incorrect because $2^x < 2x + 2$ when $0 < x < 3$ and $2^x > 2x + 2$ when $x > 3$.

▼ 44 — 0.5, 1/2

The correct answer is $\frac{1}{2}$. For the graph shown, *x* represents time, in seconds, and *y* represents momentum, in newton-seconds. Therefore, the average rate of change, in newton-seconds per second, in the momentum of the object between two *x*-values is the difference in the corresponding *y*-values divided by the difference in the *x*-values. The graph shows that at $x = 2$, the corresponding *y*-value is 6. The graph also shows that at $x = 6$, the corresponding *y*-value is 8. It follows that the average rate of change, in newton-seconds per second, from $x = 2$ to $x = 6$ is $\frac{8-6}{6-2}$, which is equivalent to $\frac{2}{4}$, or $\frac{1}{2}$. Note that 1/2 and .5 are examples of ways to enter a correct answer.

▼ 45 — A

Choice A is correct. The data show a strong linear relationship between x and y. The line of best fit for a set of data is a linear equation that minimizes the distances from the data

points to the line. An equation for the line of best fit can be written in slope-intercept form, $y = mx + b$, where m is the slope of the graph of the line and b is the y-coordinate of the y-intercept of the graph. Since, for the data shown, the y-values increase as the x-values increase, the slope of a line of best fit must be positive. The data shown lie almost in a line, so the slope can be roughly estimated using the formula for slope, $m = \frac{y_2 - y_1}{x_2 - x_1}$. The leftmost and rightmost data points have coordinates of about $(1,4)$ and $(8,26)$, so the slope is approximately $\frac{26-4}{8-1} = \frac{22}{7}$, which is a little greater than 3. Extension of the line to the left would intersect the y-axis at about $(0,1)$. Only choice A represents a line with a slope close to 3 and a y-intercept close to $(0,1)$.

Choice B is incorrect and may result from switching the slope and y-intercept. The line with a y-intercept of $(0,3)$ and a slope of 0.8 is farther from the data points than the line with a slope of 3 and a y-intercept of $(0,0.8)$. Choices C and D are incorrect. They represent lines with negative slopes, not positive slopes.

▼ 46 — B

Choice B is correct. A line in the *xy*-plane that passes through points x_1, y_1 and x_2, y_2 has a slope of $\frac{y_2 - y_1}{x_2 - x_1}$. The line of best fit shown passes approximately through the points $0, 14$ and $13, 0$. It follows that the slope of this line of best fit is approximately $\frac{0 - 14}{13 - 0}$, or $-\frac{14}{13}$. Of the given choices, -1.1 is closest to $-\frac{14}{13}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 47 — D

Choice D is correct. The data points suggest that as the variable *x* increases, the variable *y* decreases, which implies that an appropriate linear model for the data has a negative slope. The data points also show that when *x* is close to 0, *y* is greater than 9. Therefore, the *y*-intercept of the graph of an appropriate linear model has a *y*-coordinate greater than 9. The graph of an equation of the form $y = a + bx$, where *a* and *b* are constants, has a *y*-intercept

with a y-coordinate of a and has a slope of b. Of the given choices, only choice D represents a graph that has a negative slope, -0.9, and a y-intercept with a y-coordinate greater than 9, 9.4.

Choice A is incorrect. The graph of this equation has a positive slope, not a negative slope, and a y-intercept with a y-coordinate less than 1, not greater than 9.

Choice B is incorrect. The graph of this equation has a y-intercept with a y-coordinate less than 1, not greater than 9.

Choice C is incorrect. The graph of this equation has a positive slope, not a negative slope.

▼ 48 — A

Choice A is correct. It's given that the data shown in the scatterplot give the recorded temperature, in degrees Celsius, for 7 depths, in meters, below the surface of a lake. The graph of an appropriate line of best fit for the data shown passes approximately through the points $(15, 21.8)$ and $(70, 12.7)$. The slope of a line in the xy-plane that passes through the points (x_1, y_1) and (x_2, y_2) can be calculated as $\frac{y_2 - y_1}{x_2 - x_1}$. Substituting $(15, 21.8)$ and $(70, 12.7)$ for (x_1, y_1) and (x_2, y_2), respectively, in the expression $\frac{y_2 - y_1}{x_2 - x_1}$ yields $\frac{12.7 - 21.8}{70 - 15}$, or $\frac{-9.1}{55}$, which is approximately -0.165. Therefore, of the given choices, -0.17 is closest to the slope of a line of best fit for the data shown.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 49 — D

Choice D is correct. A line in the xy-plane that passes through the points x_1, y_1 and x_2, y_2 has a slope of $\frac{y_2 - y_1}{x_2 - x_1}$. The line of best fit shown passes approximately through the points $1, 3.3$ and $7, 14.5$. It follows that the slope of this best fit line is approximately $\frac{14.5 - 3.3}{7 - 1}$, which is equivalent to $\frac{11.2}{6}$, or approximately 1.87. Therefore, of the given choices, 2 is closest to the slope of the line of best fit shown.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 50 — C

Choice C is correct. Because the value of the investment increases each year, the function that best models how the value of the investment changes over time is an increasing function. It's given that each year, the value of the investment increases by 0.49% of its value the previous year. Since the value of the investment changes by a fixed percentage each year, the function that best models how the value of the investment changes over time is an exponential function. Therefore, the function that best models how the value of the investment changes over time is an increasing exponential function.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

▼ 51 — A

Choice A is correct. A line in the *xy*-plane that passes through the points (x_1, y_1) and (x_2, y_2) has a slope of $\frac{y_2 - y_1}{x_2 - x_1}$. The line of best fit shown passes through the points with approximate coordinates $(0, 3.2)$ and $(2, 4.4)$. It follows that the slope of the line of best fit is approximately $\frac{4.4 - 3.2}{2 - 0}$, which is equivalent to $\frac{1.2}{2}$, or 0.6. Therefore, of the given choices, 0.6 is closest to the slope of the line of best fit shown.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 52 — D

Choice D is correct. The relationship between the values of x and their corresponding y-values is nonlinear if the rate of change between these pairs of values isn't constant. The table for choice D gives four pairs of values: $(1,6)$,$(2,12)$, $(3,24)$, and $(4,48)$. Finding the rate of change, or slope, between $(1,6)$ and $(2,12)$ by using the slope formula, $\frac{y_2 - y_1}{x_2 - x_1}$, yields $\frac{12 - 6}{2 - 1}$, or 6. Finding the rate of change between $(2,12)$ and $(3,24)$ yields $\frac{24 - 12}{3 - 2}$, or 12.

Finding the rate of change between $(3,24)$ and $(4,48)$ yields $\frac{48-24}{4-3}$, or 24. Since the rate of change isn't constant for these pairs of values, this table shows a nonlinear relationship.

Choices A, B, and C are incorrect. The rate of change between the values of x and their corresponding y-values in each of these tables is constant, being 3, 4, and 5, respectively. Therefore, each of these tables shows a linear relationship.

▼ 53 — 6

The correct answer is 6. The line of best fit predicts a greater *y*-value than the actual *y*-value for any data point that's located below the line of best fit. For the scatterplot shown, 6 of the data points are below the line of best fit. Therefore, the line of best fit predicts a greater *y*-value than the actual *y*-value for 6 of the data points.

▼ 54 — C

Choice C is correct. It's given that the scatterplot shows the relationship between the number of seeds planted, *x*, and the number of seeds that germinated, *y*. The slope of a line of best fit represents the change in the *y*-value for a given change in the *x*-value. According to the graph, the *y*-value of the line of best fit increases by about 60 when the *x*-value increases by about 100, which represents an increase of about 60 germinated seeds for every additional 100 seeds planted. Therefore, the best interpretation of the slope of the line of best fit is that the number of seeds that germinate is predicted to increase by 60 seeds for every additional 100 seeds that are planted.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 55 — C

Choice C is correct. Any data point that's located above the line of best fit has a *y*-value that's greater than the *y*-value predicted by the line of best fit. For the scatterplot shown, 6 of the data points are above the line of best fit. Therefore, 6 of the data points have an actual *y*-value that's greater than the *y*-value predicted by the line of best fit.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the number of data points that have an actual y-value that's less than the y-value predicted by the line of best fit.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 56 — D

Choice D is correct. Each point x,y on the scatterplot gives the width x, in centimeters, and the sale price y, in dollars, of one of the 10 cutting boards put on sale at this store. According to the graph, 5 cutting boards have widths that are between 18 and 48 centimeters. These widths, in centimeters, are 24, 24, 30, 36, and 42. The sale prices, in dollars, corresponding to these widths are 10, 15, 20, 30, and 40, respectively. The average of these sale prices, in dollars, is $\frac{10+15+20+30+40}{5}$, or $\frac{115}{5}$, which is equivalent to 23. Therefore, cutting boards that have a width between 18 and 48 centimeters have an average sale price of $\$23.00$.

Choice A is incorrect. Cutting boards that have a width greater than 48 centimeters have an average sale price of $\$63.75$, not $\$59.00$.

Choice B is incorrect. Cutting boards that have a width greater than 48 centimeters have an average sale price of $\$63.75$, not $\$40.00$.

Choice C is incorrect. Cutting boards that have a width between 18 and 48 centimeters have an average sale price of $\$23.00$, not $\$31.20$.

▼ 57 — C

Choice C is correct. A line in the xy-plane that passes through the points (x_1, y_1) and (x_2, y_2) has a slope of $\frac{y_2 - y_1}{x_2 - x_1}$. The line of best fit shown passes through the points with approximate coordinates $(0, 2.5)$ and $(10, 4.7)$. It follows that the slope of the line of best fit is approximately $\frac{4.7 - 2.5}{10 - 0}$, which is equivalent to $\frac{2.2}{10}$, or 0.22. Therefore, of the given choices, 0.22 is closest to the slope of the line of best fit.

Choice A is incorrect. This is closest to the negative y-coordinate of the y-intercept, not to the slope, of the line of best fit.

Choice B is incorrect. This is closest to the negative slope, not to the slope, of the line of best fit.

Choice D is incorrect. This is closest to the y-coordinate of the y-intercept, not to the slope, of the line of best fit.

▼ 58 — 5

The correct answer is 5. For the graph shown, x represents time, in minutes, and y represents temperature, in degrees Celsius °C. Therefore, the average rate of change, in °C per minute, of the recorded temperature of the air in the chamber between two x-values is the difference in the corresponding y-values divided by the difference in the x-values. The graph shows that at $x = 5$, the corresponding y-value is 14. The graph also shows that at $x = 7$, the corresponding y-value is 24. It follows that the average rate of change, in °C per minute, from $x = 5$ to $x = 7$ is $\frac{24 - 14}{7 - 5}$, which is equivalent to $\frac{10}{2}$, or 5.

▼ 59 — C

Choice C is correct. On the line of best fit, an x-value of 25.5 corresponds to a y-value between 8 and 8.5. Therefore, at $x = 25.5$, 8.2 is closest to the y-value predicted by the line of best fit.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

▼ 60 — C

Choice C is correct. At $x = 32$, the line of best fit has a y-value between 2 and 3. The only choice with a value between 2 and 3 is choice C.

Choice A is incorrect. This is the difference between the y-value predicted by the line of best fit and the actual y-value at $x = 32$ rather than the y-value predicted by the line of best fit at $x = 32$.

Choice B is incorrect. This is the y-value predicted by the line of best fit at $x = 31$ rather than at $x = 32$.

Choice D is incorrect. This is the y-value predicted by the line of best fit at $x = 33$ rather than at $x = 32$.

▼ 61 — D

Choice D is correct. On the line of best fit, d increases from approximately 480 to 880 between $t = 12$ and $t = 24$. The slope of the line of best fit is the difference in d-values

divided by the difference in t-values, which gives $\frac{880-480}{24-12}=\frac{400}{12}$, or approximately 33. Writing the equation of the line of best fit in slope-intercept form gives $d=33t+b$, where b is the y-coordinate of the y-intercept. This equation is satisfied by all points on the line, so $d=480$ when $t=12$. Thus, $480=33(12)+b$, which is equivalent to $480=396+b$. Subtracting 396 from both sides of this equation gives $b=84$. Therefore, an equation for the line of best fit could be $d=33t+84$.

Choice A is incorrect and may result from an error in calculating the slope and misidentifying the y-coordinate of the y-intercept of the graph as the value of d at $t=10$ rather than the value of d at $t=0$. Choice B is incorrect and may result from using the smallest value of t on the graph as the slope and misidentifying the y-coordinate of the y-intercept of the graph as the value of d at $t=10$ rather than the value of d at $t=0$. Choice C is incorrect and may result from misidentifying the y-coordinate of the y-intercept as the smallest value of d on the graph.

▼ 62 — A

Choice A is correct. An equation of a line of best fit for data set F can be written in the form $y = a + bx$, where a is the y-coordinate of the y-intercept of the line of best fit and b is the slope. The line of best fit shown for data set E has a y-intercept at approximately $0, 12$. It's given that data set F is created by multiplying the y-coordinate of each data point from data set E by 3.9. It follows that a line of best fit for data set F has a y-intercept at approximately $0, 123.9$, or $0, 46.8$. Therefore, the value of a is approximately 46.8. The slope of a line that passes through points x_1, y_1 and x_2, y_2 can be calculated as $\frac{y_2 - y_1}{x_2 - x_1}$. Since the line of best fit shown for data set E passes approximately through the point $12, 30$, it follows that a line of best fit for data set F passes approximately through the point $12, 303.9$, or $12, 117$. Substituting $0, 46.8$ and $12, 117$ for x_1, y_1 and x_2, y_2, respectively, in $\frac{y_2 - y_1}{x_2 - x_1}$ yields $\frac{117 - 46.8}{12 - 0}$, which is equivalent to $\frac{70.2}{12}$, or 5.85. Therefore, the value of b is approximately 5.85, or approximately 5.9. Thus, $y = 46.8 + 5.9x$ could be an equation of a line of best fit for data set F.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This could be an equation of a line of best fit for data set E, not data set F.

▼ 63 — 4.5, 9/2

The correct answer is $\frac{9}{2}$. It's given that the scatterplot shows the relationship between the length of time y, in hours, a certain bird spent in flight and the number of days after January 11, x. Since January 13 is 2 days after January 11, it follows that January 13 corresponds to an x-value of 2 in the scatterplot. In the scatterplot, when $x = 2$, the corresponding value of y is 6. In other words, on January 13, the bird spent 6 hours in flight. Since January 15 is 4 days after January 11, it follows that January 15 corresponds to an x-value of 4 in the scatterplot. In the scatterplot, when $x = 4$, the corresponding value of y is 15. In other words, on January 15, the bird spent 15 hours in flight. Therefore, the average rate of change, in hours per day, of the length of time the bird spent in flight on January 13 to the length of time the bird spent in flight on January 15 is the difference in the length of time, in hours, the bird spent in flight divided by the difference in the number of days after January 11, or $\frac{15-6}{4-2}$, which is equivalent to $\frac{9}{2}$. Note that 9/2 and 4.5 are examples of ways to enter a correct answer.

▼ 64 — D

Choice D is correct. It's given that for $x > 0$, fx is equal to 201% of x. This is equivalent to $fx = \frac{201}{100}x$, or $fx = 2.01x$, for $x > 0$. This function indicates that as x increases, fx also increases, which means f is an increasing function. Furthermore, fx increases at a constant rate of 2.01 for each increase of x by 1. A function with a constant rate of change is linear. Thus, the function f can be described as an increasing linear function.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect. This could describe the function $fx = 2.01^x$, where fx is equal to 201% of $fx - 1$, not x, for $x > 0$.

▼ 65 — C

Choice C is correct. The predicted increase in total fat, in grams, for every increase of 1 gram in total protein is represented by the slope of the line of best fit. Any two points on the line can be used to calculate the slope of the line as the change in total fat over the change in total protein. For instance, it can be estimated that the points $(20,34)$ and $(30,48)$ are on

the line of best fit, and the slope of the line that passes through them is $\frac{48-34}{30-20}=\frac{14}{10}$, or 1.4. Of the choices given, 1.5 is the closest to the slope of the line of best fit.

Choices A, B, and D are incorrect and may be the result of incorrectly finding ordered pairs that lie on the line of best fit or of incorrectly calculating the slope.

▼ 66 — B

Choice B is correct. From the shape of the cluster of points, the line of best fit should pass roughly through the points $(1,200)$ and $(2.5,350)$. Therefore, these two points can be used to find an approximate equation for the line of best fit. The slope of this line of best fit is therefore $\frac{y_2-y_1}{x_2-x_1}=\frac{350-200}{2.5-1}$, or 100. The equation for the line of best fit, in slope-intercept form, is $y=100x+b$ for some value of b. Using the point $(1,200)$, 1 can be substituted for x and 200 can be substituted for y: $200=100(1)+b$, or $b=100$. Substituting this value into the slope-intercept form of the equation gives $y=100x+100$.

Choice A is incorrect. The line defined by $y=200x+100$ passes through the points $(1,300)$ and $(2,500)$, both of which are well above the cluster of points, so it cannot be a line of best fit. Choice C is incorrect. The line defined by $y=50x+100$ passes through the points $(1,150)$ and $(2,200)$, both of which lie at the bottom of the cluster of points, so it cannot be a line of best fit. Choice D is incorrect and may result from correctly calculating the slope of a line of best fit but incorrectly assuming the y-intercept is at $(0,0)$.

▼ 67 — A

Choice A is correct. Account A starts with $500 and earns interest at 6% per year, so in the first year Account A earns (500)(0.06) = $30, which is greater than the $25 that Account B earns that year. Compounding interest can be modeled by an increasing exponential function, so each year Account A will earn more money than it did the previous year. Therefore, each year Account A earns at least $30 in interest. Since Account B always earns $25 each year, Account A always earns more money per year than Account B.

Choices B and D are incorrect. Account A earns $30 in the first year, which is greater than the $25 Account B earns in the first year. Therefore, neither the statement that Account A always earns less money per year than Account B nor the statement that Account A earns less money than Account B at first can be true. Choice C is incorrect. Since compounding interest can be modeled by an increasing exponential function, each year Account A will earn more money than it did the previous year. Therefore, Account A always earns at least $30 per year, which is more than the $25 per year that Account B earns.

▼ 68 — A

Choice A is correct. The given equation is in slope-intercept form, or $y = mx + b$, where m is the value of the slope of the line of best fit. Therefore, the slope of the line of best fit is 0.096. From the definition of slope, it follows that an increase of 1 in the x-value corresponds to an increase of 0.096 in the y-value. Therefore, the line of best fit predicts that for each year between 1940 and 2010, the minimum wage will increase by 0.096 dollar per hour.

Choice B is incorrect and may result from using the y-coordinate of the y-intercept as the average increase, instead of the slope. Choice C is incorrect and may result from using the 10-year increments given on the x-axis to incorrectly interpret the slope of the line of best fit. Choice D is incorrect and may result from using the y-coordinate of the y-intercept as the average increase, instead of the slope, and from using the 10-year increments given on the x-axis to incorrectly interpret the slope of the line of best fit.

▼ 69 — ?

The correct answer is 285. The number of people predicted to visit the beach each day is represented by the y-values of the line of best fit, and the average temperature, in degrees Celsius ($°C$), is represented by the x-values. Since the slope of the line of best fit is approximately 57, the y-value, or the number of people predicted to visit the beach each day, increases by 57 for every x-value increase of 1, or every $1°C$ increase in average temperature. Therefore, an increase of $5°C$ in average temperature corresponds to a y-value increase of $57(5) = 285$ additional people per day predicted to visit the beach.

▼ 70 — D

Choice D is correct. The data in the scatterplot roughly fall in the shape of a downward-opening parabola; therefore, the coefficient for the x^2 term must be negative. Based on the

location of the data points, the y-intercept of the parabola should be somewhere between 740 and 760. Therefore, of the equations given, the best model is $y = -1.674x^2 + 19.76x + 745.73$.

Choices A and C are incorrect. The positive coefficient of the x^2 term means that these equations each define upward-opening parabolas, whereas a parabola that fits the data in the scatterplot must open downward. Choice B is incorrect because it defines a parabola with a y-intercept that has a negative y-coordinate, whereas a parabola that fits the data in the scatterplot must have a y-intercept with a positive y-coordinate.

▼ 71 — A

Choice A is correct. It's given that an equation for the exponential model shown can be written as $y = ab^x$, where a and b are positive constants. For an exponential model written in this form, if the value of b is greater than 0 but less than 1, the model is decreasing. If the value of b is greater than 1, the model is increasing. The exponential model shown is decreasing. Therefore, the value of b is greater than 0 but less than 1. Of the given choices, only 0.83 is a value greater than 0 but less than 1. Thus, 0.83 is closest to the value of b.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.